Score: /20

Name: _____ ID: | | | | | | | | | | |

***Read the following carefully*:**

- This is a 24-hour take-home quiz.
- Answer all **FOUR** questions. **Please show your working. If you do not show any working, no marks will be awarded even if your answer is correct.**
- Hand-write your answers. Writing with gadgets such as Apple Pencil is allowed. If you are taking photos, ensure that the photos are clear and readable.
- Save your answers as a **PDF file**, and **name it using your student ID**, for example **EEE1234567.pdf**.
- Fill in the cover page and put it as the first page of your answer script.

Question 1 [6 marks]

Given $P(A) = 0.65$, $P(B) = 0.40$ and $P(A \cap B) = 0.15$. Determine

(i) $P(A \cup B)$

(ii) $P(A^c \cap B)$

(iii) $P(B \mid A)$

(iv) If C is another event independent of B, find $P(C)$ such that $P(B \cap C) = 0.108$.

Question 2 [2 marks]

If E, F and G are mutually independent events, show that E and $(F \cap G)$ are also independent.

Question 3 [5 marks]

A true-false question is to be posed to a husband-and-wife team on a quiz show. Both the husband and the wife will independently give the correct answer with probability 0.45.

Strategy 1: Choose one of them and let that person answer the question.

Strategy 2: Have them both consider the question, and then either give the common answer if they agree or, if they disagree, flip a coin to determine which answer to give.

(i) Draw a tree diagram for Strategy 2. Clearly indicate the event labels. Write the probability on each branch.

(ii) Which of the above strategy has better winning chance for the couple?

Question 4 [7 marks]

All the lecturers in the Department of Physics like at least one of the three sports – tennis, badminton, football. There are 18 lecturers like tennis, 22 of the lecturers like badminton and 9 lecturers like football. Of all the lecturers, 10 of them like both tennis and badminton, 5 like both tennis and football, and 6 like both badminton and football. There are 4 like all 3 sports.

(i) Draw a Venn diagram. Clearly indicate the event labels and write the number of elements in the diagram.

If a lecturer is chosen randomly,

(ii) find the probability that he or she likes at least two of the sports.

(iii) find the probability that he or she likes exactly only one sport.